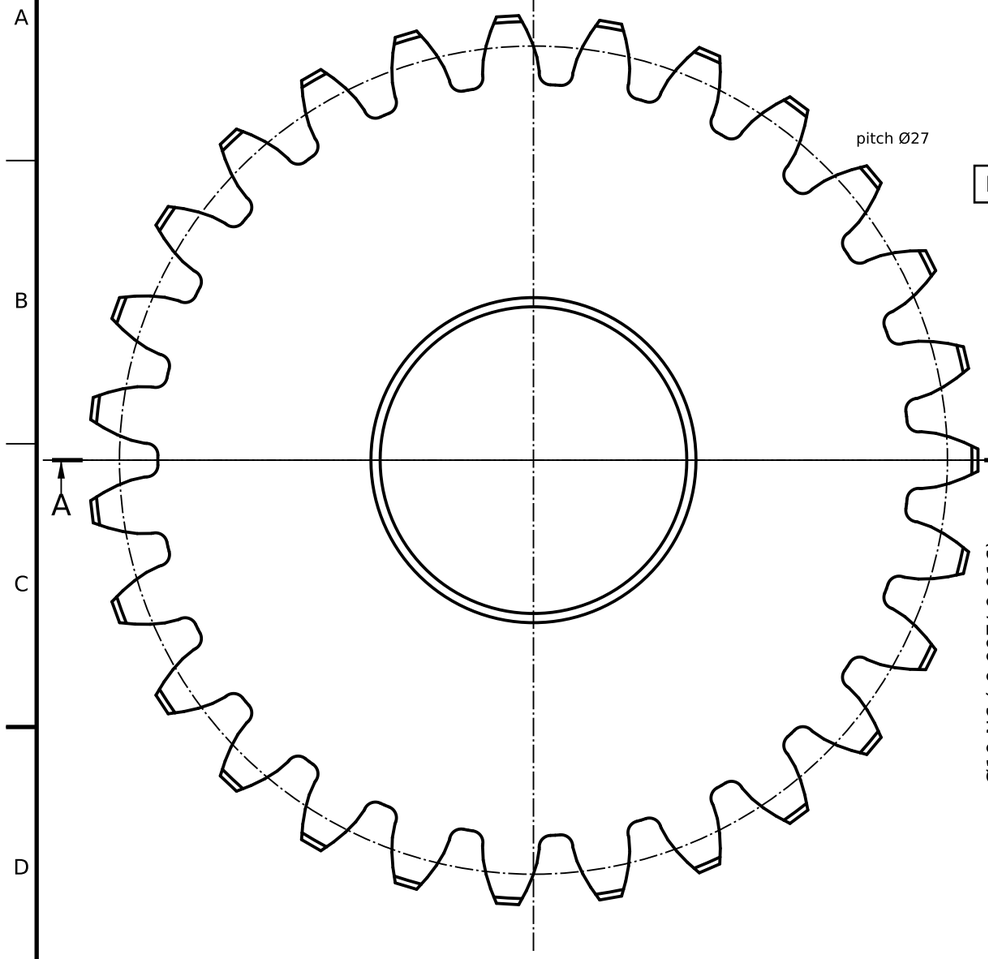
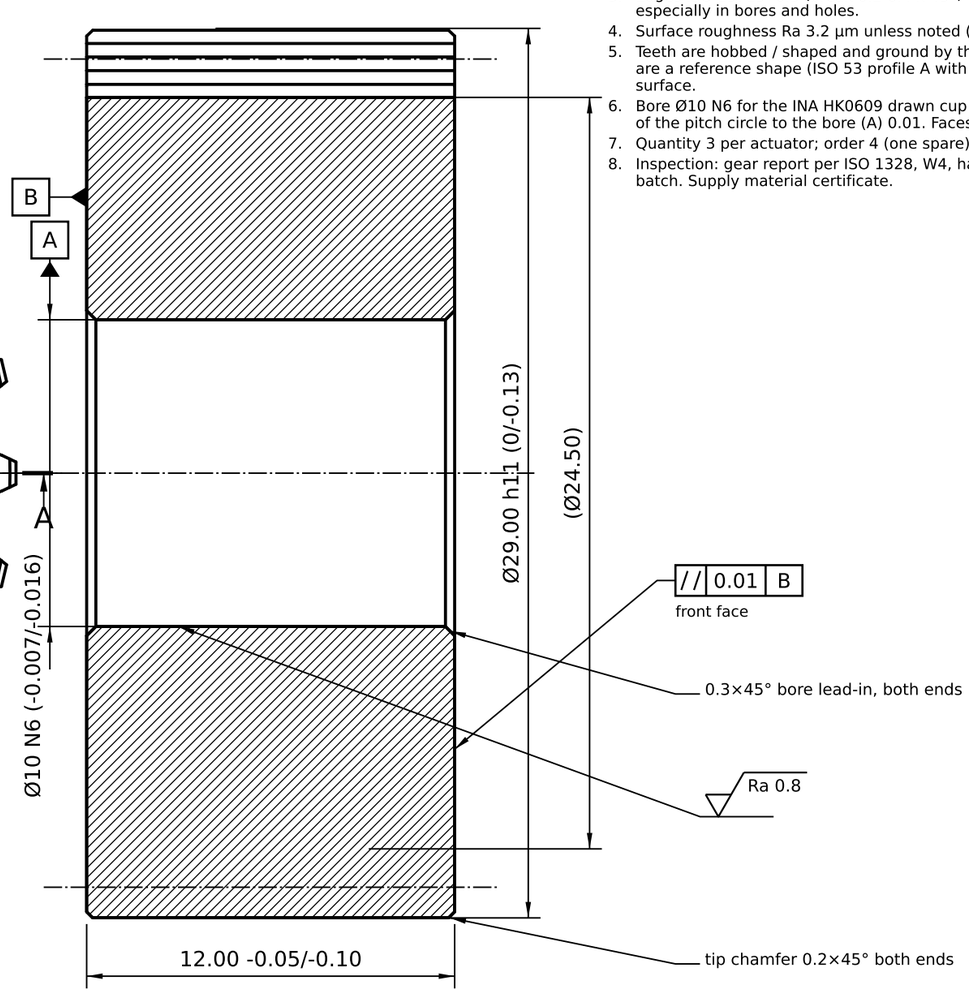
Y
X
Z

VIEW FROM FRONT (5:1)



SECTION A-A (5:1)



NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/planet.step (STEVE 02e31a10 rev 49).
3. Edges ISO 13715 -0.3 (break 0.2-0.4 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Teeth are hobbed / shaped and ground by the gear shop to the gear data table; the STEP teeth are a reference shape (ISO 53 profile A with the 0.38 m root fillet), not a machining surface.
6. Bore Ø10 N6 for the INA HK0609 drawn cup (pressed in): grind after hardening, Ra 0.8, runout of the pitch circle to the bore (A) 0.01. Faces parallel 0.01.
7. Quantity 3 per actuator; order 4 (one spare).
8. Inspection: gear report per ISO 1328, W4, hardness and case depth on a coupon from the same batch. Supply material certificate.

GEAR DATA	
Module m (normal)	1
Number of teeth z	27
Pressure angle α	20°
Helix angle β	0° (spur)
Profile shift coefficient x	0
Basic rack	ISO 53 profile A (ha* 1.0, hf* 1.25, ρf* 0.38)
Reference (pitch) diameter d	27.000
Tip diameter da	29.00 h11 (0/-0.13)
Root diameter df	24.50 (generated)
Face width b	12.00 -0.05/-0.10
Span over k = 4 teeth W4	10.711 nom.; make 10.676 / 10.656 (-0.035/-0.055)
Over 2 pins Ø1.75 Md (alt.)	29.435 nom. (odd z: pins in opposite gaps)
Working backlash	0.05-0.10 with the sun at a = 22.500
Accuracy	ISO 1328-1:2013 grade 6, teeth ground after HT
Runout of teeth (pitch circle) to A	0.01
Mating gears	sun AM-13 (z 18, a = 22.5), ring AM-15 (z 72 int.)
Material	18CrNiMo7-6 (1.6587); AISI 9310 / 8620 with cert.
Heat treatment	carburise, quench, temper
Surface hardness	58-62 HRC
Case depth	Eht 0.3-0.5 mm to 550 HV1, after grinding
Core hardness	30-42 HRC
Tooth ends	tip chamfer 0.2×45°

REV	DESCRIPTION	DATE	BY
B	Tooth thinning for 0.05-0.10 backlash (was 0.03-0.08)	2026-10-17	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash	SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 49; STEP: cnc/step/planet.step			
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081); Edges ISO 13715 -0.3 (0.2-0.4)	PROJECTION: FIRST ANGLE, ISO 128 / ISO 5456-2				
FINISH / HEAT TREATMENT: Carburised 58-62 HRC, Eht 0.3-0.5, ground				UNITS: mm	
MATERIAL: 18CrNiMo7-6 (1.6587)				SCALE: 5:1	
TITLE: **Planet gear m1 27 T**				QTY PER ACTUATOR: **3 (+1 spare)**	
DRAWING / PART NUMBER: **AM-14-planet**	REV: **B**	SHEET: 1/1	SIZE: A3	DATE OF ISSUE: 2026-10-17	